Autoencoder and Adversarial-Learning-Based Semisupervised Background Estimation for Hyperspectral Anomaly Detection

Weiying Xie, *Member, IEEE*, Baozhu Liu, Yunsong Li, Jie Lei, Qian Du, *Fellow, IEEE*

***Abstract*— Reliable detection of anomalies without any prior information is a critical yet challenging task in many applications, not least military and civilian fields. An intelligent anomaly detection system would use the material-specific spectral information in hyperspectral images (HSIs), thereby avoiding the loss of visually confusing objects. However, conventional hyperspectral anomaly detection methods are mainly achieved in an unsupervised way leading to limited performance due to lack of prior knowledge. In this article, we propose a novel autoencoder and adversarial-learning based semisupervised background estimation model (SBEM) that is trained only on the background spectral samples in order to accurately learn the background distribution. In particular, an unsupervised background searching method is firstly conducted on the original HSIs to search the background spectral samples. Our proposed SBEM consists of an encoder, a decoder, and a discriminator to thoroughly capture background distribution. Furthermore, jointly minimizing the reconstruction loss, spectral loss, and adversarial loss during training aids the model to learn the background distribution as required. Experiments on four real HSIs demonstrate that compared to the current state-of-the-art, the proposed framework yields higher detection capability and lower false alarm rate, which shows that it has a significant benefit in the tradeoff between detection accuracy and false alarm rate.**

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Weiying Xie, Baozhu Liu, and Yunsong Li are with the State Key Laboratory of Integrated Service Networks, Xidian University, Xi'an 710071, China (e-mail: wyxie@xidian.edu.cn; bzliu_1@stu.xidian.edu.cn; ysli@mail.xidian.edu.cn).

Jie Lei is with the State Key Laboratory of Integrated Service Networks, Xidian University, Xi'an 710071, China, and also with the Science and Technology on Electrooptic Control Laboratory, Luoyang 471000, China (e-mail: jielei@mail.xidian.edu.cn).

Qian Du is with the Department of Electronic and Computer Engineering, Mississippi State University, Starkville, MS 39759 USA (e-mail: du@ece.msstate.edu).

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***Index Terms*— Anomaly detection, background searching and estimation, hyperspectral image (HSI), semisupervised learning.**

I. Introduction

HYPERSPECTRAL remotely sensed images are of great interest because their high spectral resolution can be regarded as a well-suited measure for many remote sensing applications, such as land cover classification [1], [2] and anomaly detection [3], [4]. As an essential task in these applications, anomaly detection aims to locate spectral instances that deviate from the background. Anomalies are defined as samples that appear significantly different spatially or spectrally from their surrounding background samples. Anomaly detection cannot be simply treated as a binary classification in an unsupervised way because the occurrence of anomalies is much lower than that of background [5].

Various methods for anomaly detection have emerged in recent years. Among them, statistics-based methods are the most well-known to distinguish anomalies from the background via the leverage of their statistic differences. In particular, Reed Xiaoli (RX) algorithm proposed by Reed and Yu [6] is viewed as a milestone approach that first built the background statistical model by a multivariate Gaussian distribution. Subsequently, the test pixels are determined to be anomalies or not by estimating the Mahalanobis distance between the test pixels and the model constructed. Numerous variants based on the RX detector, such as the local RX detector (LRX) [7], the global RX (GRX) [7], and the kernel RX (KRX) [8], have been proposed to improve detection performance. Unfortunately, the RX-based methods generally calculate the background mean and covariance matrix on a local region or an entire image scene that are inevitably contaminated or disturbed by anomalies.

With the development of the sparsity theory, considerable progress has been made in hyperspectral anomaly detection. Collaborative-representation-based detector (CRD) [9] is one of the algorithms based on collaborative representation, which exploits the assumption that the background pixel can be roughly represented by the linear combination of its spatial neighborhoods while anomaly cannot. Background joint sparse representation (BJSR) algorithm employs an unsupervised adaptive subspace model to suppress the background and simultaneously highlight anomaly components [10]. In particular, the adaptive orthogonal background complementary

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subspace is evaluated by the BJSR. Based on the observation that the background in hyperspectral images (HSIs) is low-rank and anomalies are sparse, a novel anomaly detection method is proposed in [11] by modeling the background via a low-rank representation and determining anomalies via representation residuals. Different from the aforementioned works, Li *et al.* [12] proposed an attribute and edge-preserving filtering-based anomaly detection (AED) method, which outperforms many state-of-the-art approaches in terms of computing time and detection accuracy.

Recently, deep learning-based methods have attracted great attention, and a considerable volume of algorithms based on deep learning for hyperspectral anomaly detection has been proposed [13]–[16]. Because of the unknown prior information, there is almost no supervised method for anomaly detection. In practical applications, unsupervised methods have become a significant research topic owing to the fact that these methods allow less dependence on the label information for the training data set. Most of the unsupervised methods attempt to learn the deep representation of HSIs for better anomaly detection performance [17]. A stacked autoencoders (AEs)-based adaptive subspace model (SAEASM) is proposed in [13] to extract the in-depth features of differences for calculating the detection results. Xie *et al.* [18] proposed a spectral constrained adversarial AE (SC_AAE) network to obtain more discriminative representation of HSIs, which can suppress the background samples while preserving the characteristic of anomalies. The success of unsupervised AE-based or AAE-based methods is supported by their potent capacity to draw high-dimensional probability distributions of the inputs without any prior information [19]. However, it is a challenge of high false alarm and low detection accuracy due to the lack of prior knowledge of anomalies with respect to unsupervised methods. Anomaly methods based on semisupervised have been popularly used in many domains, for example, the hybrid semisupervised anomaly detection model consists of a deep AE (DAE) and an ensemble k-nearest neighbor graphs-(K-NNG-) based anomaly detector is proposed in [20] for high-dimensional data. Only normal samples are available for training in semisupervised methods. Next, in the testing process, test samples are regarded as anomalies when their difference is far between normal samples. In many practical applications, such as network monitoring [21], lesion diagnosis based on medical images [22], normal samples are easy to obtain for semisupervised training.

To approach the performance of supervised method while relaxing the lack of availability of labeled training samples, we propose a semisupervised background estimation model (SBEM) for hyperspectral anomaly detection, which only leverages background samples to train a model, and the model is then applied to all the samples. Based on the observation that anomalies in HSIs are difficult to be defined or searched because they appear at an extremely low probability and may show minor diversification due to the spectral reflection of the surrounding pixels, the SBEM-based hyperspectral anomaly detection method firstly searches the background samples that are easier to search than anomalies. Then, it can obtain the background estimation function based on the searched background samples. Finally, all the spectral samples are fed into the background estimation function with the result of more discriminative representation between background samples and anomalies. Compared with other anomaly detection algorithms, the major contribution of our proposed method is summarized as follows.

1) An unsupervised clustering method is first performed to extract a relatively pure background where pixels with a high probability of belonging to background are remained, on the contrary, those with a high probability of belonging to anomaly are rejected. To estimate the background distribution, it is best to first reject anomalies as not belonging to any of the training background samples.
2) In particular, we design a semisupervised background estimation network for jointly learning reconstruction loss, spectral loss, and adversarial loss to capture the distribution of the searched background samples. As far as we know, it is the first attempt to construct an anomaly detection model by a semisupervised measure in hyperspectral anomaly detection fields.
3) The estimated HSIs can improve the difference between the anomalies and the background. Experiments on the data sets containing both point and structural anomalies demonstrate that our method can maintain detection capability with a lower false alarm rate.

This article is organized as follows. Section II is a brief introduction to the related work. Section III presents the proposed method in detail. Section IV provides experimental results. Finally, Section V concludes this article.

II. Related Work

A. Autoencoder

As a self-taught learning neural network, AE, composed of an encoder and a decoder, forces the outputs $\hat{\mathbf{x}}$ to equal its inputs $\mathbf{x}$ by minimizing the reconstruction loss between $\hat{\mathbf{x}}$ and $\mathbf{x}$. The encoder in AE first projects the inputs $\mathbf{x}$ into the latent space $\mathbf{z}$ by using the function $\varphi(\cdot) : \mathbf{x} \rightarrow \mathbf{z}$, while the decoder attempts to reconstruct the hidden code $\mathbf{z}$ by using the function $\varphi'(\cdot) : \mathbf{z} \rightarrow \hat{\mathbf{x}}$ to guarantee the outputs $\hat{\mathbf{x}}$ to preserve all details of the inputs $\mathbf{x}$. Mapping functions $\varphi(\cdot)$ and $\varphi'(\cdot)$ indicate the nonlinear activation functions to provide nonlinearity for the neural network. There are many alternatives for the activation function, such as sigmoid, hyperbolic tangent, and ReLU function. In AE, the sigmoid function is chosen. The objective of AE is to minimize the error between the input and the reconstruction one

$$L = \|\mathbf{x} - \hat{\mathbf{x}}\|_2. \tag{1}$$

The outputs of AE is only determined by the information of the hidden layer activity, which is extracted from the input. Therefore, the latent space $\mathbf{z}$ is usually seen as a low-dimensional feature representation of the input with high-dimensions. When AE converges, the learned feature in the latent layer is obtained by removing the decoder and its parameters, which is generally applied for anomaly detection, classification [18].

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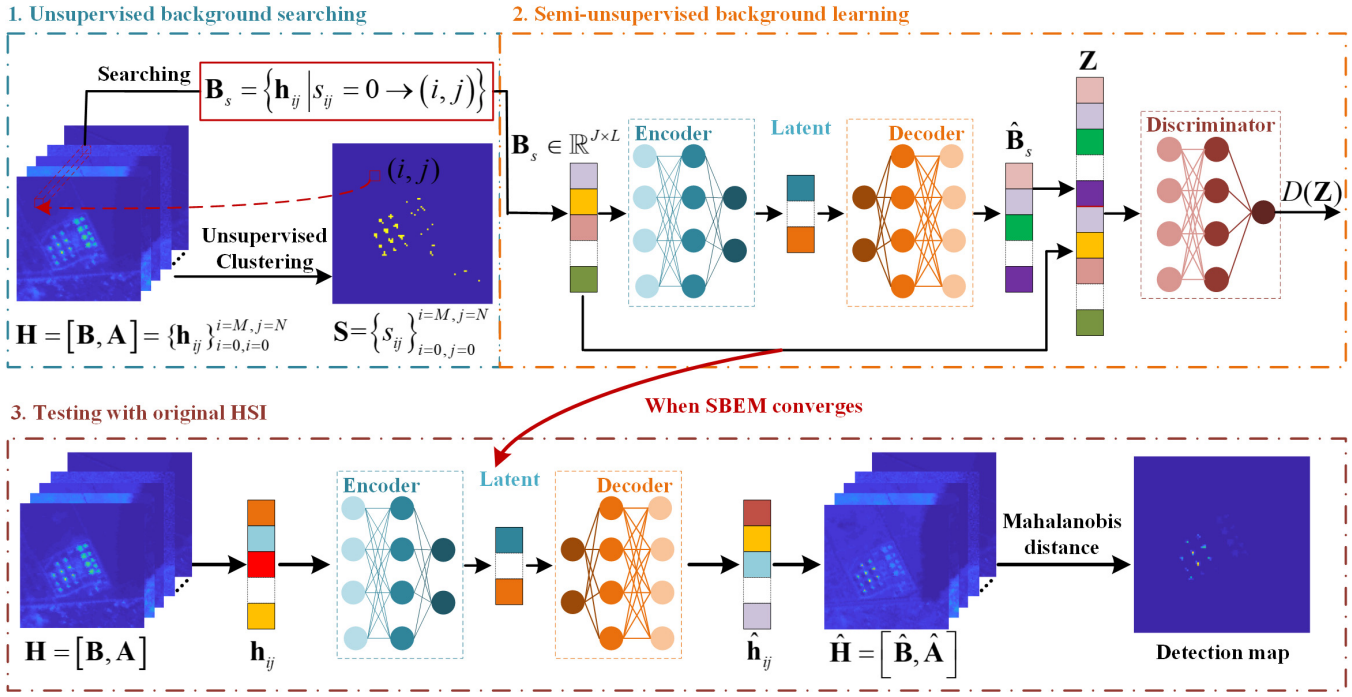


Fig. 1. Schematic of the proposed anomaly detection framework in HSIs.

B. Generative Adversarial Network

Generative adversarial network (GAN) proposed in [23] is a popular framework to evaluate generative models via an adversarial process, of which a generator G and a discriminator D are composed. A min–max adversarial game is played between the two sub-networks during the training process. The discriminator D calculates the probability of the sample $\mathbf{x}$ that comes from the real data we feed into the model rather than the generative model by using the function $D(\mathbf{x})$. Currently, the generator transforms the noise $\mathbf{n}$ into data space via the generative function $D(\mathbf{n})$. During the training process, $G(\mathbf{n})$ is responsible for maximally cheating the discriminator so that D is convinced of the data that G produces arises from the real data distribution. Afterward, $D(\mathbf{x})$ attempts to distinguish the real data distribution from the fake data created by the generator. In other words, the training can be treated as a procedure to optimize the following function:

$$\min_G \max_D \mathbf{V}(G, D) = E[\log D(\mathbf{x})] + E[\log(1 - D(G(\mathbf{n})))]. \tag{2}$$

The capacity to model complex high-dimensional data distributions, especially image distribution, allows GAN to earn the most popularity and approval in extensive research. Particularly, the power of GAN to generate data with state-of-the-art performance has enabled it to open new possibilities for anomaly detection [24], [25].

III. Semisupervised Background Estimation Model for Hyperspectral Anomaly Detection

In this section, we construct a novel framework for hyperspectral anomaly detection, which is based on the fact that anomalies in HSIs are rare or never present and are substantially different from the known background. From another perspective, our proposed SBEM aims to detect anomalies that do not conform to the characteristics of background, which is composed of three major components: unsupervised background searching, semiunsupervised background learning, and testing with original HSI. Fig. 1 shows the schematic of the proposed SBEM-based hyperspectral anomaly detection. It is expected to facilitate modeling of the background without anomaly contamination.

Before formally describing the framework, the notations used in this article are first introduced. An HSI $\mathbf{H} \in \mathbb{R}^{M \times N \times L}$ contains background $\mathbf{B}$ and anomaly $\mathbf{A}$, i.e., $\mathbf{H} = [\mathbf{B}, \mathbf{A}] = \{\mathbf{h}_{ij}\}_{i=0,j=0}^{i=M,j=N}$, where $M \times N$ and L denote the number of pixels in spatial domain and bands, respectively. Here, $\mathbf{h}_{ij}$ represents the spectral vector whose spatial position is (i, j). All spectral vectors $\mathbf{h}_{ij}$ are collected into matrix form $\mathbf{H}$. With these notations, the goal of the proposed SBEM-based hyperspectral anomaly detection is to find a predict function $\mathbf{\Phi}$ such that

$$\hat{\mathbf{h}}_{ij} = \mathbf{\Phi}(\mathbf{h}_{ij}; \mathbf{B}_s | \mathbf{B}_s \in \mathbf{B}). \tag{3}$$

The function $\mathbf{\Phi}$ aims at modeling background distribution while failing to reconstruct anomalies. Therefore, the reconstructed $\hat{\mathbf{H}}$ has more distinguishing features to identify anomalies from the background in comparison to the original $\mathbf{H}$. In an ideal case, $\mathbf{B}_s$ is completely searched from background $\mathbf{B}$, which is one of the challenges. Besides, it is significant to design $\mathbf{\Phi}$ to learn background distribution from the searched subset $\mathbf{B}_s$ in the proposed framework.

A. Unsupervised Background Searching

It is important to search a subset $\mathbf{B}_s$ for further training background estimation model. It is worth noting that

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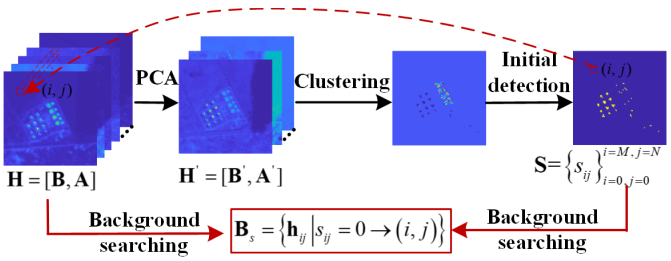


Fig. 2. Procedure of initially rejecting anomalies and retaining background in HSIs.

background searching should be implemented in an unsupervised manner. Thus, it is difficult to find clean background vectors with anomalies rejected initially. Considering the simple yet powerful observations that anomalies occur with low probability, it is to design a searching method that obtains a clean background where pixels with high probability of belonging to background remained, on the contrary, those with high probability of belonging to anomaly are rejected. In other words, the design of the searching method can be regarded as a problem of separating background and anomalies initially.

Here, density-based spatial clustering of applications with noise (DBSCAN) [26] algorithm is employed to reject anomalies initially, which is capable of spotting clusters of different sizes and shapes with no need to specify the number of clusters in advance. It first defines clusters as the separation of high-density areas and low-density ones through the use of a density estimator. Afterward, DBSCAN groups all mutually density-connected points to compose one cluster while marking points that individually exist in the low-density areas as noise. Note that points characterized as noise are also referred to as anomalies [27]. Hence, DBSCAN is naturally able to achieve anomaly detection [27]–[29], which is the major difference compared to other algorithms.

Anomalies in HSIs are also in the low-density areas so that DBSCAN is applied as the key piece of the coarse filter to reject anomalies roughly and retain background in HSIs. The detailed framework of the initial detection and background searching is shown in Fig. 2. It is well known that DBSCAN is susceptible to high-dimensional data, as it turns out that the complexity of DBSCAN depends on the dimension of the input data. To address this problem, principal component analysis (PCA) [30] is applied to feature extraction and dimension reduction of HSIs. Thus, in this article, PCA is combined with DBSCAN to improve the effectiveness of DBSCAN [27]. Firstly, PCA is introduced to reduce the dimension of input HSIs while keeping important information. Note that we set the parameter of PCA to 0.99 to obtain the top 99% principal components of the input data. Subsequently, DBSCAN is applied in the transformed HSIs to find anomalies. Concretely, the procedure of initial detection and background searching is presented in Algorithm 1.

Clustering maps can be derived by utilizing the clustering ability of DBSCAN. Here, we focus on the separation of anomalies and background rather than a detailed classification of objects in HSIs. Therefore, classes that are not specified as anomalies by DBSCAN are collectively classified into one class, which can be regarded as background. Assign background class to 0, while anomalies to 1 to acquire the initial detection $\mathbf{S}$. For the given input data matrix $\mathbf{H}$, we can take advantage of $\mathbf{S}$ to generate a relatively pure background-based data $\mathbf{B}_s$ by selecting points with the value of 0 and rejecting the samples with a high rate of being anomalies, which is represented as

$$\mathbf{B}_s = \{\mathbf{h}_{ij} | s_{ij} = 0 \rightarrow (i, j)\}. \tag{4}$$

As discussed in our model (3), once the background samples $\mathbf{B}_s$ is precisely determined for learning background distribution, the function $\mathbf{\Phi}$ needs to be designed to model background distribution, which is discussed in Section III-B.

Algorithm 1 Realization of the Background Searching

Input: original HSI $\mathbf{H} = [\mathbf{B}, \mathbf{A}]$, maximum distance between two points in the same group $\mathcal{E}$, minimum number of samples in a cluster *MinPts*;

Steps:

1: Reduce the dimension of input data $\mathbf{H}$ by PCA and set the transformed data as $\mathbf{H}' = [\mathbf{B}', \mathbf{A}'] = \{\mathbf{h}_{ij}\}_{i=0,j=0}^{i=M',j=N'}$, where $M' < M$, $N' < N$;

2: Calculate the clusters $\mathbb{C}_1, \mathbb{C}_2 \cdots \mathbb{C}_c$ and the anomalies $\mathbb{Q}$ by applying DBSCAN algorithm.

3: Find the corresponding class of each vector $\mathbf{h}'_{ij}$ in $\mathbf{H}'$ according to $\mathbb{C}_1, \mathbb{C}_2 \cdots \mathbb{C}_c$ and $\mathbb{Q}$, and display all classes as a clustering map as shown in Fig. 2;

4: Set anomalies in clustering map to 1, while other background classes to 0 for obtaining the initial detection $\mathbf{S}$;

5: Search background vector from original input $\mathbf{H}$ according to Eq. (4).

Output: background samples $\mathbf{B}_s$.

B. Semiunsupervised Background Learning

It is well known that semisupervised learning for anomaly detection can be described as the identification of normal (training) observation and anomaly (testing) observation. In other words, the semiunsupervised anomaly detection task aims only to leverage the normal samples for training neural networks. In this article, we pay more attention to the problem of the background spectral distribution estimation with the assumption of background spectral vectors being considered as normal observation in HSIs. The learned SBEM is able to generate realistic-looking spectral vectors while input is normal, whereas to synthesize a low-quality estimation when input is anomalous. As mentioned above, the background vectors $\mathbf{B}_s$ sampled from $\mathbf{B}$ are utilized as the training spectral vectors for SBEM to learn background spectral characteristic.

1) Network Architecture: As is illustrated in Fig. 1, the proposed SBEM consists of three subneural networks: an encoder, a decoder, and a discriminator. The AE consisting of encoder and decoder is responsible for reconstructing the input data firstly. For a given background spectral data $\mathbf{B}_s$, the AE

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creates a new data $\hat{\mathbf{B}}_s$ by utilizing the mapping function $\hat{\mathbf{B}}_s = An(\mathbf{B}_s)$, This learning of $An(\cdot)$ can be decomposed into two stages. Firstly, encoder reflects input background vectors as the hidden representation $\mathbf{I}$ with the encoding process $f(\cdot)$ (5). The decoder then employs the decoding process $g(\cdot)$ (6) to reconstruct $\mathbf{B}_s$ into $\hat{\mathbf{B}}_s$ according to the hidden code $\mathbf{I}$. Hence, the background spectral vectors encoding process can be expressed as

$$\mathbf{I} = f(\mathbf{W}^e \mathbf{B}_s + \delta^e) \tag{5}$$

$$\hat{\mathbf{B}}_s = g(\mathbf{W}^d \mathbf{I} + \delta^d) \tag{6}$$

where, the parameters $\mathbf{W}^e$ and δ^e signify the weight matrix and the bias vector of the background spectral vectors encoding layers, whereas the parameters $\mathbf{W}^d$ and δ^d denote the weight matrix and the bias term of the background spectral vectors decoding layers.

The training object of an AE is to minimize the deviation between a reconstruction from the initial value as much as possible. Generally, the Euclidean distance or the mean squared error (MSE) is the loss function in most of the AE-based methods and is applied to quantify the deviation of the reconstructed data from the original input data. The original reconstruction loss of an AE is employed to ensure the performance of the neural network, which is defined as

$$L_r = \|\mathbf{B}_s - \hat{\mathbf{B}}_s\|_2. \tag{7}$$

However, Euclidean (l2) norm or MSE is easy to cause the produced image to be blurry [31]. Considering the inherent spectral characteristics of spectral vectors, we employ spectral distance (SD) that is based on the spectral angle mapper (SAM) to constrain the input and renewed spectral vectors. SAM [32], which is used to estimate the spectral similarity between two input vectors, is applied as an evaluation index in many fields [33], [34]. The SD with the total number of J vectors in $\mathbf{B}_s$ is defined as

$$\mathrm{SD} = \left(1 - \frac{1}{\pi} \arccos\left(\frac{\mathbf{B}_s \hat{\mathbf{B}}_s}{\|\mathbf{B}_s\|_2 \|\hat{\mathbf{B}}_s\|_2}\right)\right). \tag{8}$$

The value of SD is in the range of [0,1], and the greater the value, the more similar the two input vectors are. It is adopted as a term of the loss function named as spectral loss to optimize the generator for the sake of taking the spatial and spectral feature information into account. Such a mechanism is executed to ensure that the generator reconstructs better and converges faster. Therefore, the complete loss function is

$$L_A = \|\mathbf{B}_s - \hat{\mathbf{B}}_s\|_2 - \left(1 - \frac{1}{\pi} \arccos\left(\frac{\mathbf{B}_s \hat{\mathbf{B}}_s}{\|\mathbf{B}_s\|_2 \|\hat{\mathbf{B}}_s\|_2}\right)\right). \tag{9}$$

In the adversarial learning stage, a GAN composed of the AE and the discriminator is trained for the purpose of improving the quality of generated data. As mentioned above, the AE, also called as generator G, aims to create new data fed into the discriminator D. In particular, another input of discriminator is the input background spectral vector $\mathbf{B}_s$. The output of the discriminator is a single scalar value, which indicates the probability that $\mathbf{B}_s$ arises from the initial data rather than the generated data $\hat{\mathbf{B}}_s$. D is trained to maximize the probability to distinguish the label of $\mathbf{B}_s$ and $\hat{\mathbf{B}}_s$. Simultaneously, G is trained to minimize the $\log(1 - D(An(\mathbf{B}_s)))$ for producing samples $An(\mathbf{B}_s)$ that obeys the distribution of $\mathbf{B}_s$ and fool discriminator D. The two-player min–max game between G and D can be described by the following:

$$L_{GD} = \min_G \max_D E[\log D(\mathbf{B}_s)] + E[\log(1 - D(An(\mathbf{B}_s)))]. \tag{10}$$

Moreover, encoder, decoder, and discriminator in the SBEM model all contain two fully connected layers with 500 units in each layer. The size of latent space lying between encoder and decoder is restricted to 20 units. The leaky ReLU (LReLU) is a linear function to compress the negative input with a predefined constant parameter in the range (0, 1), which can retain the negative information compared to ReLU activation function. So all the layers in encoder, decoder, and discriminator employ LReLU as an activation function except that the last layer of an encoder (having no activation) and the last layer of a decoder (utilizing tanh activation), and the last layer of discriminator (using sigmoid activation). Due to limited training samples, it is easy to cause overfitting during training. Batch normalization (BN) proposed in [35] can regularize networks without parameter adjustment in many fields. The strategy is used to avoid overfitting and its effectiveness of speeding up the convergence has been demonstrated. In order to decrease iteration epochs and stabilize the training process, BN is immediately added before the LReLU in the network.

2) Training Process: The SBEM network is similar to adversarial AE (AAE), which is introduced in [36]. However, with respect to the network structure, SBEM model is different from the AAE in essence. AAE aims to learn a generative model based on the input data by uniting the reconstruction error with an adversarial training criterion with a desired arbitrary prior distribution being imposed on the latent representation [37]. The encoder in AAE is adopted to convert the input data distribution to an imposed prior distribution, i.e., Gaussian distribution, whereas the decoder learns to map the specified prior to the data distribution. On the other hand, the discriminator fixed between encoder and decoder is trained to discriminatively distinguish whether the input generated by the encoder or specified by users. Because of its representational capability, AAE can extract intrinsic features of the given data [38].

Nevertheless, semiunsupervised anomaly detection is dedicated to addressing the issue of how to represent background data distribution. Thus, less attention is focused on the quality of the extracted features compared to the reconstruction capabilities of the network. Inspired by the combination of the reconstruction and the adversarial training in AAE, we employ the discriminator to differentiate the input data arise from the original data or reconstructed by the decoder. In spite of the motivation from AAE, the frame of SBEM resembles a GAN to a greater extent, wherein the encoder and the decoder performed as a generator. In particular, the distribution of the latent is not constrained.

The training process of the SBEM consists of two phases: the reconstruction stage and the adversarial stage. First, in the

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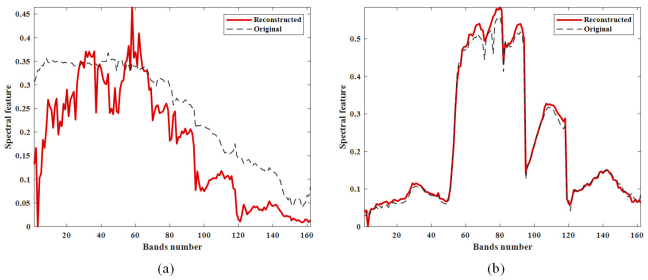


Fig. 3. Comparison of the reconstructed and original spectral feature in HYDICE. (a) Anomaly. (b) Background.

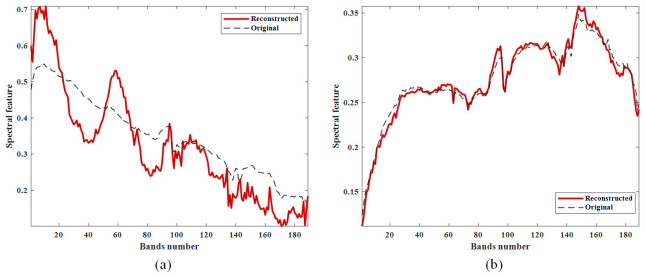


Fig. 4. Ccomparison of the reconstructed and original spectral feature in San Diego. (a) Anomaly. (b) Background.

reconstruction stage, an AE composed of the encoder and the decoder attempts to minimize the reconstruction error and spectral loss by updating its training parameters. In the adversarial stage, the AE operates as a generator and forms a GAN with the discriminator. The discriminator followed predicts the discriminative probability that a sample is from the real data (positive samples) that are fed into the SBEM rather than a sample produced by the generator (negative samples). Concurrently, the generator modifies its parameters to maximally deceive the discriminator for making it believe that the samples it generates are the real data distribution. These two steps are performed alternately per minibatch under the guidance of minimizing reconstruction error and the classification error.

When the encoder, decoder, and discriminator are jointly learned for thousands of epochs until the SBEM converges the parameters $(\mathbf{W}^e, \delta^e, \mathbf{W}^d, \delta^d)$ of function $\mathbf{\Phi}$ are fixed and used to reconstruct the original HSI.

C. Testing With Original HSI

For a given $\mathbf{H} = [\mathbf{B}, \mathbf{A}] = \{\mathbf{h}_{ij}\}_{i=0,j=0}^{i=M,j=N}$, it is reconstructed by the trained function $\mathbf{\Phi}$ with parameters $(\mathbf{W}^e, \delta^e, \mathbf{W}^d, \delta^d)$, that is

$$\hat{\mathbf{h}}_{ij} = \mathbf{\Phi}(\mathbf{h}_{ij}|\mathbf{W}^e, \delta^e, \mathbf{W}^d, \delta^d). \tag{11}$$

The proposed SBEM aims to facilitate modeling of the background without contamination by anomalies, which is learned on the background samples $\mathbf{B}_s$. Therefore, if a spectral vector $\mathbf{h}_{ij}$ comes from background $\mathbf{B}$, it will be better reconstructed by the background estimation function $\mathbf{\Phi}$, otherwise, estimation of $\mathbf{h}_{ij}$ fails when it comes from $\mathbf{A}$. Figs. 3 and 4 display the difference of original and reconstructed spectral vectors on anomalous pixels and background pixels from two different data sets. Obviously, anomalies in HSIs are not well recovered, whereas background pixels are reconstructed with less reconstruction error. It is evident that background and anomalies are more divergent through (11). Finally, the Mahalanobis distance between the test sample and the background is calculated to determine whether the pixel is an anomaly or not.

IV. Experimental Results

In this section, the anomaly detection performance of our proposed approach is demonstrated by four commonly used real hyperspectral data sets. These four data sets captured at different scenes containing different types of anomalies are introduced as follows.

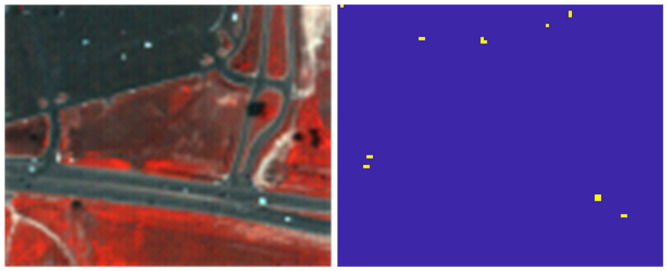
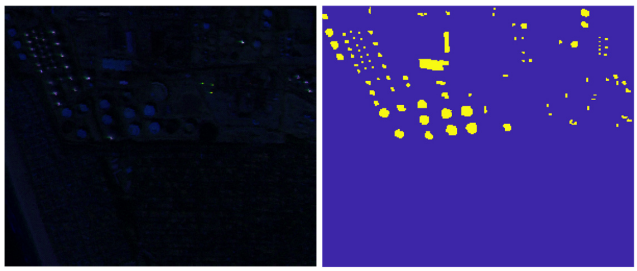


Fig. 5. (a) Visualization and (b) corresponding groundtruth maps of HYDICE.

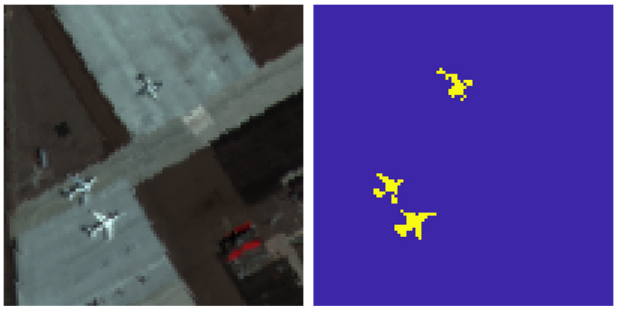
1) *HYDICE Data Set:* This data set describes an urban scene in the USA, which was captured by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) airborne sensor. The original image of size 307 × 307, with 162 spectral channels ranging from 400 to 2500 nm. A subimage of the spatial size 80 × 100 is cropped from the whole original image for hyperspectral anomaly detection. As for the spectral bands, 162 bands are preserved for retaining useful information while other bands are removed for eliminating redundant information. The total anomalous pixels are 19 and account for 0.2375%, which are cars and roofs. The color composite of this data set and its corresponding groundtruth are shown in Fig. 5(a) and (b), respectively.
2) *EI Segundo Data Set:* The EI Segundo collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor scans an area that includes an oil refinery, few residential areas, parks, and a school zone. The data set, including 224 spectral channels in the range 366–2496 nm, has a spatial size of 250 × 300 with a ground resolution of 7.1 m. The total anomalous pixels are 2048, accounting for 2.7307% of the total number of pixels in this data set. The false-color image and its corresponding groundtruth map of this data set are shown in Fig. 6(a) and (b), respectively.
3) *San Diego Data set:* The third real hyperspectral data set was also captured by the AVIRIS sensor, which has a spatial resolution of 3.5 m and a spectral resolution

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(a) (b)

Fig. 6. (a) Visualization and (b) corresponding groundtruth maps of EI Segundo.



(a) (b)

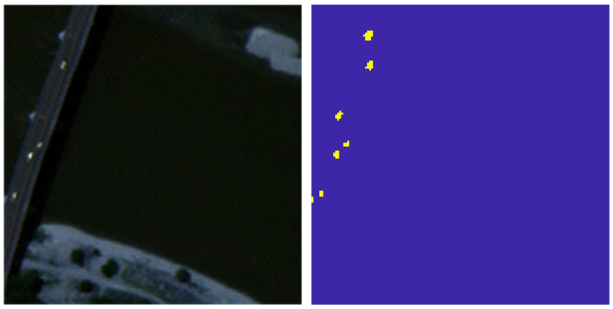
Fig. 7. (a) Visualization and (a) corresponding groundtruth maps of San Diego.

of 10 nm. The data set covers the San Diego airport area, CA, USA, and 134 pixels in three airports embedded in the background are regarded as anomalies to be detected. The original spatial size of the data set is 100 × 100, and the number of spectral bands is 224. After eliminating several noisy bands (1–6, 33–35, 97, 107–113,153–166, and 221–224) by considering the water vapor and atmospheric effects, the other 189 bands are preserved. The visualization picture and its corresponding groundtruth map of this AVIRIS data set are shown in Fig. 7(a) and (b), respectively.

4) *Pavia Data Set:* The fourth data acquired by the Reflective Optics System Imaging Spectrometer sensor shows the scene of Pavia city. This scene consists of 150 × 150 pixels, with 175 spectral channels in wavelengths ranging from 430 to 860 nm. With bridge and water as the main background, vehicles on the bridge of 68 target pixels are to be detected. Furthermore, the spatial resolution is approximately 1.3 m. The sample image and the groundtruth map for this data set are shown in Fig. 8(a) and (b), respectively.

A. Evaluation Criteria

The receiver operating characteristic (ROC) curve and the area under the ROC curve (AUC) are the most commonly used criteria for intuitively evaluating the performance of anomaly detection. The ROC curve of (P_d, P_f) can effectively illustrate the relationship between the true-positive rate (P_d) and false-positive rate (P_f) by plotting these two rates at a set of



(a) (b)

Fig. 8. (a) Visualization and (a) corresponding groundtruth maps of Pavia.

threshold settings (τ). An anomaly detector performs better when the probability of detection is higher than the other ones at the same false alarm by observing the ROC curve. Similarly, the AUC score of (P_d, P_f) is a metric to evaluate detection performance quantitatively. It is known that the closer the AUC score of (P_d, P_f) is to 1, the more outstanding the detector is. Furthermore, the AUC score of (P_f, τ) indicates the area value under the curve encircled by P_f and τ, which reports the probability of false detection for a detector. When the value of the false alarm is 0, it means no background pixel is mistakenly detected as an anomaly.

B. Comparison and Parameter Setup

To demonstrate the effectiveness of the proposed method, five state-of-the-art algorithms including AED [12], LSMAD [39], RX [6], LRX [7], and CRD [9] are compared with our SBEM-based method. As leading performance detectors, they are known as the benchmark in the realm of hyperspectral anomaly detection and are frequently cited in different literature. Hence, we compare the detection maps, AUC scores of (P_d, P_f) and (P_f, τ) to evaluate the detection performance both visually and quantitatively.

The parameters in our proposed method and the other compared detectors are set for optimal detection performance to each data set rely on the corresponding AUC scores of (P_d, P_f). For the proposed SBEM-based method, the depth of the encoder, decoder, and discriminator in the SBEM network are set to be three layers, respectively. Furthermore, these three layers are claimed to be fully connected layers with 500 units in each of the first two layers and 20 units in the last layer. For encoder and decoder, the learning rate is restricted to 0.001, while for the discriminator it is 0.0001. Nevertheless, parameters in the SBEM network can be tuned for other different data sets to obtain better performance. Parameters of all comparative methods in this article are set optimal. For AED, it is known that the area parameter ∂ mainly determines the accuracy of detection. In particular, in order to acquire the optimal results, the range of ∂ is set between 5 and 300 for different data sets. For LSMAD, the detection capacity is correlated with the rank value γ, for which we select different rank values for each data set by an empirical method. For LRX and CRD, the size of the inner window w_i and outer window w_o are two significant parameters highly related to

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TABLE I
PARAMETERS SETTING FOR AED, LSMAD, LRX, AND CRD ON THE EXPERIMENT DATA SETS

	AED	LSMAD	LRX		CRD	
	∂	γ	w_o	w_i	w_o	w_i
HYDICE	5	4	5×5	3×3	15×15	9×9
EI Segundo	300	4	31×31	9×9	31×31	13×13
San Diego	25	8	101×101	78×78	51×51	39×39
Pavia	5	4	7×7	5×5	15×15	13×13

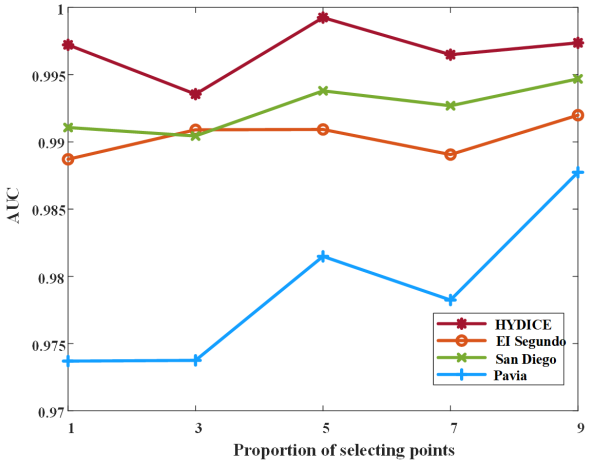


Fig. 9. Sensitivity analysis of the proportion of selecting points on different data sets.

the performance of detection. Table I describes the detailed parameter setting in this article for the compared algorithms.

C. Parameter Analysis

1) Proportion of Selecting Points: In this section, we analyze the effect of the proportion ϑ of selection background pixels from the original HSIs. According to the aforementioned description, we can search for background pixels in HSIs depending on the result of the unsupervised clustering. However, due to the existence of extremely few anomalies, the selected background pixels may not be completely pure. Thus, more selected background pixels cannot guarantee better performance. We test the proportion of selected points in $\mathbf{B}_s$ as a series of numbers: 1, 1/3, 1/5,1/7,1/9. Fig. 9 shows the influence of ϑ over the AUC scores of (P_d, P_f) of the SBEM-based method on each experimented data set. It can be observed that when the rate equals 1/9 almost all data sets obtained optimal detection accuracy except for the HYDICE data set. For HYDICE, it can be seen that when $\vartheta = 1/5$, the AUC score of (P_d, P_f) is best, which is only a little higher than that acquired when $\vartheta = 1/9$. In other words, it is considered that when $\vartheta = 1/9$, the proposed method can offer satisfactory detection performance. Hence, we set the parameter ϑ as 1/9.

2) Number of Hidden Nodes: The number of hidden nodes eta in the SBEM is a vital parameter, which greatly impacts

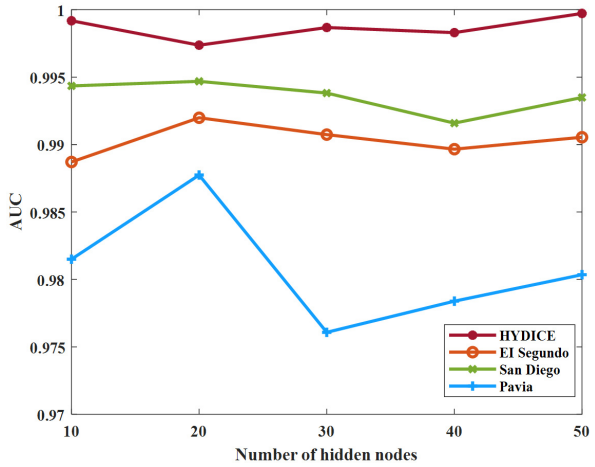


Fig. 10. Number of hidden nodes sensitivity analysis on different data sets.

TABLE II
EVALUATION AUC SCORES OF (P_d, P_f) OBTAINED BY THE SBEM, AED,LSMAD, RX, LRX, AND CRD

Data Set	SBEM	AED	LSMAD	RX	LRX	CRD
HYDICE	**0.9974**	0.9899	0.9968	0.9764	0.9920	0.9621
EI Segundo	**0.9920**	0.9801	0.9786	0.9815	0.9444	0.9343
San Diego	**0.9947**	0.9850	0.9830	0.9515	0.9467	0.9805
Pavia	**0.9877**	0.9794	0.9620	0.9534	0.9244	0.9322
Average	**0.9929**	0.9836	0.9801	0.9657	0.9519	0.9523

on the learning ability of the network. We set η to 10, 20, 30, 40, 50 and assess the AUC scores of (P_d, P_f) on each experimented data set. As shown in Fig. 10, for almost all data sets, the AUC scores of (P_d, P_f) are the best when the number of hidden nodes is chosen as 20 except for HYDICE data set, which obtained its optimal value when $\eta = 10$. In a nutshell, in order to simplify the parameter setting in this article, the default parameter of η is set to 20 for all data sets.

D. Detection Results

Tables II and III show the AUC scores of (P_d, P_f) and (P_f, τ) of different methods on the four real HSIs. It can be seen that the proposed method presents competitive performances in terms of the AUC scores of (P_d, P_f). Meanwhile, for the sake of further investigating the detection performance quantitatively, the corresponding ROC curves of (P_d, P_f) and (P_f, τ) are introduced. Figs. 11 and 12 show the ROC curves of (P_d, P_f), and (P_f, τ) on each data set, respectively. It is noticeable that the SBEM is outstanding due to the fact that all the ROC curves of (P_d, P_f) lie nearer the upper left corner, which indicates that the SBEM is capable of obtaining a high probability of detection with a low false alarm rate. Although the LSMAD and LRX methods demonstrate their superiority of suppressing the interference of background

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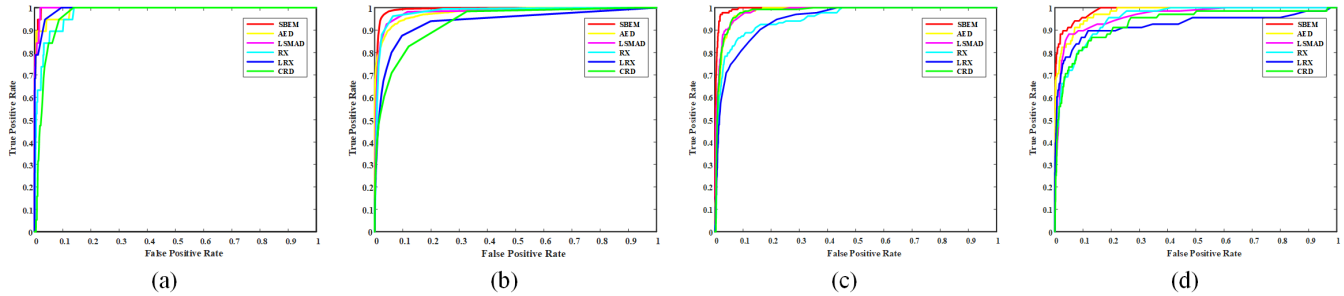


Fig. 11. ROC curves of (P_d, P_f) of the algorithms for (a) HYDICE, (b) EI Segundo, (c) San Diego, and (d) Pavia.

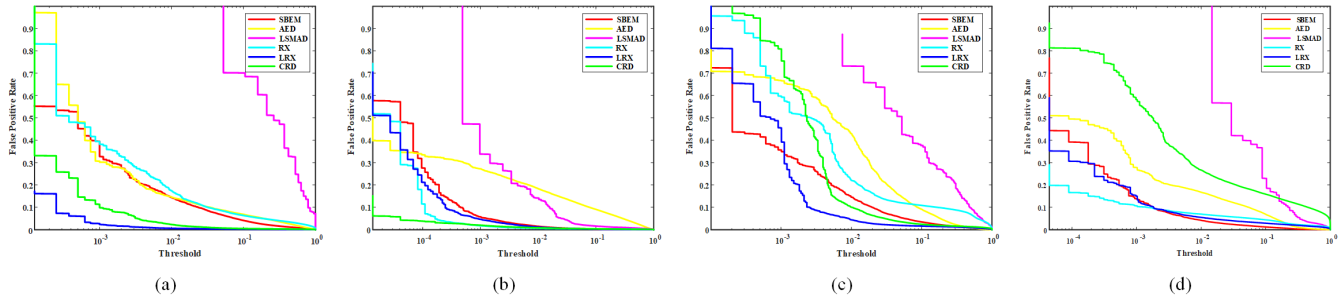


Fig. 12. ROC curves of (P_f, τ) of the algorithms for (a) HYDICE, (b) EI Segundo, (c) San Diego, and (d) Pavia.

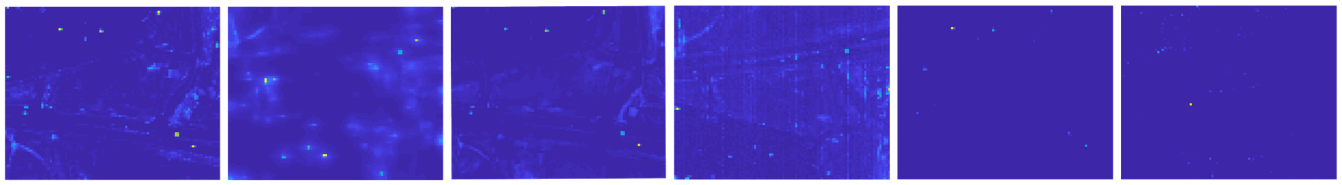


Fig. 13. Visualization of the detection results on HYDICE. (a) SBEM. (b) AED. (c) LSMAD. (d) RX. (e) LRX. (f) CRD.

TABLE III
EVALUATION AUC SCORES OF (P_f, τ) OBTAINED BY THE SBEM, AED, LSMAD, RX, LRX, AND CRD

Data Set	SBEM	AED	LSMAD	RX	LRX	CRD
HYDICE	0.0184	0.0309	0.0159	0.0380	**0.0040**	0.0047
EI Segundo	0.0010	0.0393	**0.0007**	0.0017	0.0015	0.0021
San Diego	0.0162	0.0333	0.0102	0.0704	**0.0050**	0.0587
Pavia	**0.0044**	0.0200	0.0050	0.0233	0.0194	0.1124
Average	0.0100	0.0309	0.0079	0.0333	**0.0075**	0.0445

around anomalies, they are unable to find the location of anomalies with high intensity. The SBEM lower than AED in the false alarm rate and higher in the detection accuracy, which further illustrates its excellent performance of the SBEM-based method.

For HYDICE data set, the detection maps are described in Fig. 13. It can be observed that the proposed method can effectively identify anomalies with 0.9974 as the AUC score of (P_d, P_f) and these detected anomalies show higher response compared with other algorithms. The detection performance of AED and RX is poor because of the lower AUC scores of (P_d, P_f) and higher AUC scores of (P_f, τ). The performance of LSMAD is close to the proposed method in terms of the detection performance. However, the anomalies are almost invisible in its detection map. The LRX and CRD methods obtain a better separation between anomalous pixels and the background, while they are unable to detect more anomalies accurately.

Fig. 14 shows the detection maps for EI Segundo data set. It is apparent that the proposed method can accurately distinguish anomalies and background with the highest AUC score of (P_d, P_f) of 0.9920 and a lower AUC score of (P_f, τ), which implies its superiority. For AED, it fails to separate anomalous pixels and surrounding background so that it cannot overcome the interference of the complex relatively background. The LSMAD, RX, LRX, and CRD methods successfully retain the background at the expense of missing part of anomalies.

For San Diego data set, the detection maps are presented in Fig. 15. From these maps, it is obvious that the SBEM can locate the position of three airplanes with full shapes

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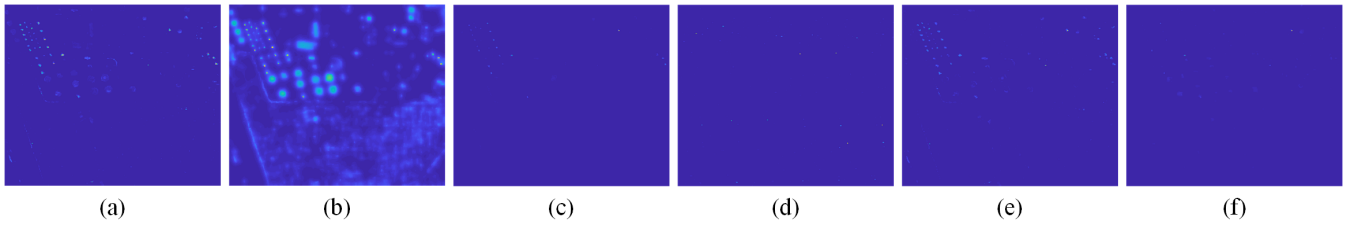


Fig. 14. Visualization of the detection results on EI Segundo. (a) SBEM. (b) AED. (c) LSMAD. (d) RX. (e) LRX. (f) CRD.

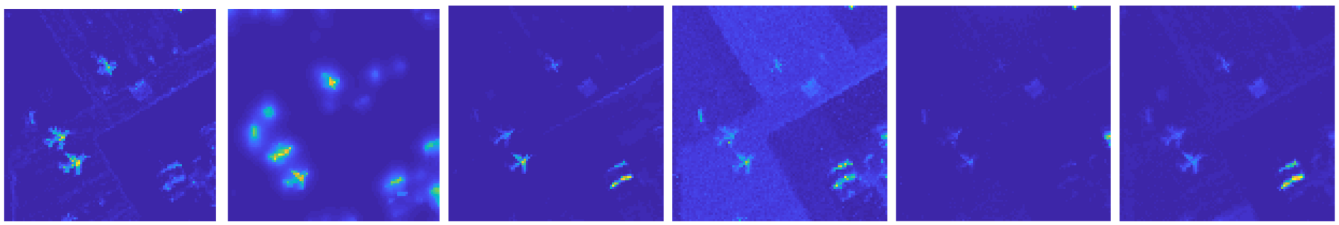


Fig. 15. Visualization of the detection results on San Diego. (a) SBEM. (b) AED. (c) LSMAD. (d) RX. (e) LRX. (f) CRD.

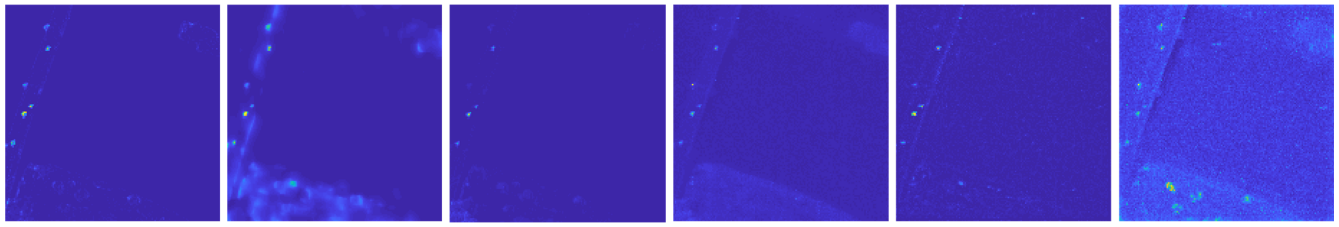


Fig. 16. Visualization of the detection results on Ravia. (a) SBEM. (b) AED. (c) LSMAD. (d) RX. (e) LRX. (f) CRD.

are detected. The AUC score of (P_d, P_f) is 0.9947, which is the best result compared to the other methods. Nevertheless, the AED and LSMAD methods can also generate pretty good detection results, while the shapes of airplanes are fuzzy and missed. Moreover, the contrast between airplanes and background are low while the CRD keeps targets relatively clearly. The RX and LRX methods fail to detect all the anomalies, leading to lower AUC scores of (P_d, P_f).

For Pavia data set, the detection maps are shown in Fig. 16. It is apparent that the AUC score of (P_d, P_f) is 0.9887 and the AUC score of (P_f, τ) is 0.0044, which shows the proposed method outperforms the other four comparison methods. The detection map obtained by the SBEM yields a larger separation between anomalies and background. Moreover, all targets in the bridge are detected clearly. The detection map of the LSMAD method seems to be less influenced by the background, but anomalies are also suppressed and invisible. The AED can identify almost all anomalies, but the background is mistakenly considered as anomalies. In addition, the RX, LRX, and CRD miss most targets, generating lower detection accuracy.

Besides, to further support the necessity of combining adversarial-learning with AE in the proposed method, the additional comparison experiments based on original AE-based anomaly detection are conducted. We utilize AE instead of SBEM during the comparison experiments to detect anomalies in these four real HSIs. The AUC scores of (P_d, P_f), and (P_f, τ) of AE-based anomaly detection are shown in Table IV. It is obvious that the AUC scores of (P_d, P_f) of AE-based method are smaller than the proposed method and the average AUC score of (P_f, τ) of AE-based method is higher than the proposed method, which certifies that the combination of adversarial-learning with AE in the proposed method can contribute to improving the detection performance.

Based on the above analysis, it can be proved that the detection performance of our proposed method is outstanding compared to the state-of-the-art anomaly detection methods.

TABLE IV
EVALUATION AUC SCORES OF (P_d, P_f) AND (P_f, τ) OF THE AE-BASED METHOD

Data Set	(P_d, P_f)	(P_d, τ)
HYDICE	0.9959	0.0271
EI Segundo	0.9721	0.0005
San Diego	0.9882	0.0164
Pavia	0.9715	0.0043
Average	0.9819	0.0121

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V. Conclusion

In this article, we propose a novel AE and adversarial-learning-based anomaly detection method named SBEM in HSIs. It is composed of three complementary workflows: unsupervised background searching, semisupervised background learning, and testing with the original HSI. In unsupervised background searching, we exploit the observations that anomalies have low occurrence probability and are substantially different from the known background. This process ensures that a clean background where pixels with a high probability of belonging to the background remains. In semisupervised background learning, reconstruction loss, spectral loss, and adversarial loss are jointly trained to capture background distribution as required thoroughly. This process constructs the background estimation function based on the searched background samples. Finally, all the spectral samples are fed into the background estimation function. Experimental results on four real hyperspectral data sets show that the proposed SBEM is superior to the other anomaly detection methods. Its superior performance validates the importance of semisupervised learning in hyperspectral anomaly detection. In the future, we will tune all of the hyperparameters so as to see the largest performance gain of the proposed model for the specific data set.

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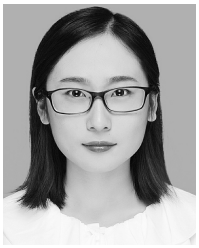
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Weiying Xie (Member, IEEE) received the B.S. degree in electronic information science and technology from the University of Jinan, Jinan, China, in 2011, the M.S. degree in communication and information systems from the Lanzhou University, Lanzhou, China, in 2014, and the Ph.D. degree in communication and information systems from the Xidian University, Xi'an, China, in 2017.

She is currently an Associate Professor with the State Key Laboratory of Integrated Services Networks, Xidian University. She has authored or coauthored more than 20 articles in refereed journals, including the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, *Neural Networks*, and PATTERN RECOGNITION. Her research interests include neural networks, machine learning, hyperspectral image processing, and high-performance computing.



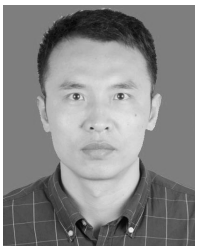
Baozhu Liu received the B.E. degree in telecommunications engineering from the Xidian University, Xi'an, China, in 2018, where she is currently pursuing the M.S. degree with the Image Coding and Processing Center, State Key Laboratory of Integrated Service Network.

Her research interests include hyperspectral image processing, anomaly detection, and machine learning.



Yunsong Li received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from the Xidian University, Xi'an, China, in 1999 and 2002, respectively.

He joined the School of Tetecommunications Engineering, Xidian Uniersity, in 1999, where he is currently a Professor. He is also the Director of the Image Coding and Processing Center, State Key Laboratory of Integrated Service Networks, Xidian Uniersity. His research interests focus on image and video processing and high-performance computing.



Jie Lei received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from the Xidian University, Xi'an, China, in 2006 and 2010, respectively.

He was a Visiting Scholar with the Department of Computer Science, University of California at Los Angeles, Los Angeles, CA, USA, from 2014 to 2015. He is currently an Associate Professor with the School of Telecommunications Engineering, Xidian University. He is also a member of the Image Coding and Processing Center, State Key Laboratory of Integrated Services Networks, and also with the Science and Technology on Electrooptic Control Laboratory, Luoyang, China. His research interests focus on image and video processing, computer vision, and customized computing for big-data applications.



Qian Du (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland, Baltimore, MD, USA, in 2000.

She is currently the Bobby Shackouls Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. She is also an Adjunct Professor with the College of Surveying and Geo-informatics, Tongji University, Shanghai, China. Her research interests include hyperspectral remote sensing image analysis and applications, pattern classification, data compression, and neural networks.

Dr. Du is a fellow of the SPIE-International Society for Optics and Photonics. She received the 2010 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society. She was the Co-Chair of the Data Fusion Technical Committee of the IEEE Geoscience and Remote Sensing Society from 2009 to 2013, and the Chair of the Remote Sensing and Mapping Technical Committee of the International Association for Pattern Recognition from 2010 to 2014. She has served as an Associate Editor of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the *Journal of Applied Remote Sensing*, and the IEEE SIGNAL PROCESSING LETTERS. Since 2016, she has been the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. She was the General Chair of the fourth IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing held at Shanghai, China, in 2012.

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